entitled **"Experiments on Plant Hybrids"** was presented and published in The Proceedings of the Brunn Society of Natural History in 1866. Mendel was the first systematic researcher in the field of genetics.

Mendel was successful because:

- He applied mathematics and statistical methods to biology and laws of probability to his breeding experiments.
- He followed scientific methods and kept accurate and detailed records that include quantitative data of the outcome of his crosses.
- His experiments were carefully planned and he used large samples.
- The pairs of contrasting characters which were controlled by factor (genes)were present on separate chromosomes. (Figure 2.4)
- The parents selected by Mendel were pure breed lines and the purity was tested by self crossing the progeny for many generations.

Mendel's Experimental System – The Garden pea.

He chose pea plant because,

- It is an annual plant and has clear contrasting characters that are controlled by a single gene separately.
- Self-fertilization occurred under normal conditions in garden pea plants. Mendel used both self-fertilization and cross-fertilization.
- The flowers are large hence emasculation and pollination are very easy for hybridization.

2.2.2 Mendel's experiments on pea plant

Mendel's theory of inheritance, known as the Particulate theory, establishes the existence of minute particles or hereditary units or factors, which are now called as **genes**. He performed artificial pollination or cross pollination experiments with several true-breeding lines of pea plants. A true breeding lines (Pure-breeding strains) means it has undergone continuous self pollination having stable trait inheritance from parent to offspring. Matings within pure breeding lines produce offsprings having specific parental traits that are constant in inheritance and expression for many generations. Pure line breed refers to homozygosity only. Fusion of male and female gametes produced by the same individual i.e pollen and egg are

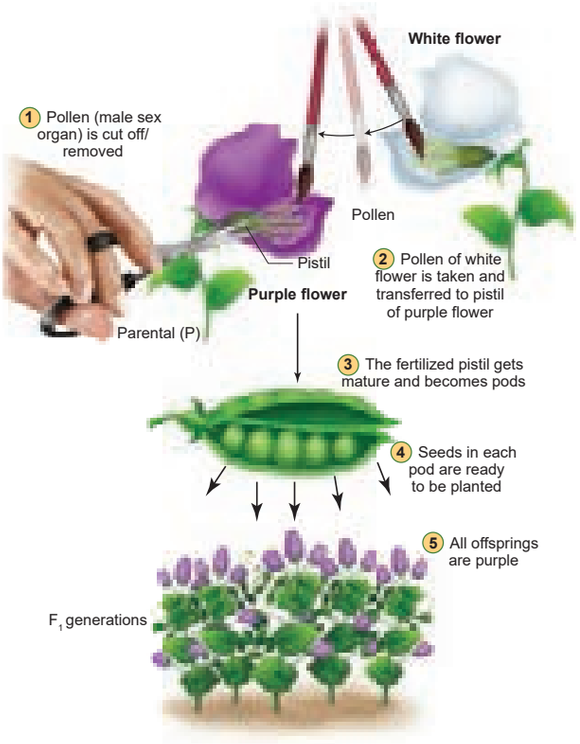


Figure 2.2: Steps in cross pollination of pea flowers

derived from the same plant is known as self-fertilization. Self pollination takes place in Mendel's peas. The experimenter can remove the anthers (Emasculation) before fertilization and transfer the pollen from another variety of pea to the stigma of flowers where the anthers are removed. This results in cross-fertilization, which leads to the creation of hybrid varieties with different traits. Mendel's work on the study of the pattern of inheritance and the principles or laws formulated, now constitute the Mendelian Genetics.

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